

Evaluation of Tracking Systematic Uncertainties via Roadset Variance

Step-by-Step Methodology for the Drell-Yan Absolute Cross-Section

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July 20, 2026

Motivation: Tracking Systematics

Objective: Quantify the systematic uncertainty introduced by different trigger road definitions (RS57, RS59, RS62, RS67, RS70).

- **Challenge:** Different roadsets have drastically different Protons on Target (POT) and differing trigger acceptances.
- **Approach:** Calculate the cross-section independently for each roadset, then evaluate the bin-by-bin variance.
- **Constraint:** Because the reliability of high-statistics roadsets (e.g., RS67) is under evaluation, we must treat roadsets equally to prevent biased pulls.

Step 1 & 2: Independent Execution & Dynamic Normalization

To prevent normalization distortions (the "Global POT Trap"), the analysis pipeline was refactored to execute *per roadset*.

Each execution dynamically overrides the global POT with the roadset-specific POT to yield the true independent cross-section:

Roadset	LH2 POT	LD2 POT	Flask POT
RS57	3.53×10^{16}	1.76×10^{16}	3.92×10^{15}
RS59	9.37×10^{15}	4.32×10^{15}	1.01×10^{15}
RS62	5.28×10^{16}	2.38×10^{16}	1.10×10^{16}
RS67	1.61×10^{17}	7.69×10^{16}	3.66×10^{16}
RS70	1.79×10^{16}	8.75×10^{15}	3.84×10^{15}

Step 3: Variance Calculation Mathematics

For a given kinematic bin, let x_i be the absolute cross-section for roadset i , and e_i be its statistical error.

Inverse-Variance Weighting:

To suppress massive Poisson fluctuations from low-POT datasets (e.g., RS59), we scale by statistical power ($w_i = 1/e_i^2$):

$$\mu_w = \frac{\sum w_i x_i}{\sum w_i}$$

$$\sigma_{\text{sys},w} = \sqrt{\frac{\sum w_i}{(\sum w_i)^2 - \sum w_i^2} \sum w_i (x_i - \mu_w)^2}$$

Step 4: Resulting Uncertainties (1D p_T - LH2)

Using the inverse-variance weighted strategy suppresses low-statistics noise, yielding the following systematic errors for Liquid Hydrogen.

Note: Bins with zero relative error or relative error $\geq 50\%$ have been omitted.

p_T Bin	Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
0	[0.00, 0.32)	0.001794	0.000290	16.18%
1	[0.32, 0.49)	0.003872	0.000511	13.19%
2	[0.49, 0.63)	0.004383	0.000536	12.24%
3	[0.63, 0.77)	0.004766	0.000651	13.65%
4	[0.77, 0.95)	0.004313	0.000533	12.37%
5	[0.95, 1.18)	0.003721	0.000662	17.79%
6	[1.18, 1.80)	0.001853	0.000248	13.40%

Step 4: Resulting Uncertainties (1D p_T - LD2)

Applying the same inverse-variance weighted procedure to Liquid Deuterium targets yields comparable systematic margins.

Note: Bins with zero relative error or relative error $\geq 50\%$ have been omitted.

p_T Bin	Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
0	[0.00, 0.32)	0.003968	0.000698	17.59%
1	[0.32, 0.49)	0.008777	0.001154	13.15%
2	[0.49, 0.63)	0.010581	0.001340	12.67%
3	[0.63, 0.77)	0.010224	0.001331	13.02%
4	[0.77, 0.95)	0.010045	0.001534	15.27%
5	[0.95, 1.18)	0.007935	0.001482	18.67%
6	[1.18, 1.80)	0.004107	0.000738	17.98%

Double Differential Systematics (xF and Mass)

Extension to 2D Kinematics:

- The inverse-variance weighting procedure is extended to the double differential bins (xF and Invariant Mass).
- The following slides document the systematic uncertainties strictly for statistically significant data sets across both LH2 and LD2 targets.
- **Filtering Criteria:** Bins reporting a relative variance $\leq 0\%$ or $\geq 50\%$ represent zero-bin artifacts or severe low-statistics starvation and have been omitted for clarity.

Double Differential Systematics (xF and Mass) - LH2 (1/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.00, 0.05)	[4.70, 4.80)	1.6875e+00	8.0930e-01	47.96%
[0.00, 0.05)	[4.80, 4.90)	5.3545e-01	1.1833e-03	0.22%
[0.00, 0.05)	[5.00, 5.10)	7.0517e-01	2.2411e-01	31.78%
[0.00, 0.05)	[5.10, 5.20)	3.1273e-01	1.3585e-01	43.44%
[0.00, 0.05)	[5.50, 5.60)	2.8546e-01	6.4372e-02	22.55%
[0.00, 0.05)	[5.60, 5.70)	2.5180e-01	9.9260e-02	39.42%
[0.00, 0.05)	[6.30, 6.40)	1.0091e-01	4.3884e-02	43.49%
[0.00, 0.05)	[6.40, 6.50)	9.3820e-02	3.0568e-02	32.58%
[0.00, 0.05)	[6.70, 6.80)	1.1296e-01	5.2463e-02	46.44%
[0.00, 0.05)	[7.40, 7.50)	7.9309e-02	1.2638e-03	1.59%
[0.05, 0.10)	[4.50, 4.60)	6.1000e-01	2.7131e-01	44.48%

Double Differential Systematics (xF and Mass) - LH2 (2/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.05, 0.10)	[5.20, 5.30)	3.7091e-01	1.4889e-01	40.14%
[0.05, 0.10)	[5.50, 5.60)	3.2507e-01	3.4217e-02	10.53%
[0.05, 0.10)	[5.90, 6.00)	1.4632e-01	4.0668e-02	27.79%
[0.05, 0.10)	[6.00, 6.10)	1.3115e-01	4.0076e-02	30.56%
[0.05, 0.10)	[6.70, 6.80)	2.9629e-02	3.1893e-03	10.76%
[0.05, 0.10)	[6.80, 6.90)	6.5397e-02	3.2990e-03	5.04%
[0.05, 0.10)	[7.20, 7.30)	4.8263e-02	1.2623e-02	26.15%
[0.05, 0.10)	[7.50, 7.60)	7.2150e-02	5.8238e-03	8.07%
[0.10, 0.15)	[4.30, 4.40)	7.9932e-01	2.7357e-01	34.22%
[0.10, 0.15)	[4.50, 4.60)	6.9059e-01	3.3090e-01	47.92%
[0.10, 0.15)	[4.60, 4.70)	7.0544e-01	1.5247e-01	21.61%

Double Differential Systematics (xF and Mass) - LH2 (3/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.10, 0.15)	[5.10, 5.20)	4.2855e-01	1.7372e-01	40.54%
[0.10, 0.15)	[5.30, 5.40)	3.2284e-01	6.3484e-02	19.66%
[0.10, 0.15)	[5.40, 5.50)	2.0275e-01	6.6870e-02	32.98%
[0.10, 0.15)	[5.50, 5.60)	2.6983e-01	4.3891e-02	16.27%
[0.10, 0.15)	[5.60, 5.70)	2.2670e-01	8.3990e-02	37.05%
[0.10, 0.15)	[5.80, 5.90)	1.9682e-01	5.3040e-02	26.95%
[0.10, 0.15)	[6.10, 6.20)	1.3328e-01	5.8612e-02	43.98%
[0.10, 0.15)	[6.20, 6.30)	1.4490e-01	5.7711e-02	39.83%
[0.10, 0.15)	[6.30, 6.40)	8.9151e-02	2.5787e-02	28.93%
[0.10, 0.15)	[6.50, 6.60)	7.3539e-02	2.1519e-02	29.26%
[0.10, 0.15)	[7.00, 7.10)	7.5465e-02	2.8594e-02	37.89%

Double Differential Systematics (xF and Mass) - LH2 (4/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.15, 0.20)	[4.30, 4.40)	1.4193e+00	6.3634e-01	44.83%
[0.15, 0.20)	[4.50, 4.60)	6.2577e-01	2.8094e-01	44.90%
[0.15, 0.20)	[4.70, 4.80)	9.3163e-01	1.3228e-01	14.20%
[0.15, 0.20)	[4.90, 5.00)	6.8903e-01	1.3576e-01	19.70%
[0.15, 0.20)	[5.40, 5.50)	2.9419e-01	6.9339e-02	23.57%
[0.15, 0.20)	[5.50, 5.60)	2.0453e-01	8.9809e-02	43.91%
[0.15, 0.20)	[5.60, 5.70)	1.4787e-01	6.3790e-02	43.14%
[0.15, 0.20)	[5.80, 5.90)	1.7906e-01	5.7233e-02	31.96%
[0.15, 0.20)	[5.90, 6.00)	1.7475e-01	7.3061e-02	41.81%
[0.15, 0.20)	[6.00, 6.10)	1.3315e-01	2.1119e-02	15.86%
[0.15, 0.20)	[6.20, 6.30)	1.1274e-01	4.7839e-02	42.43%

Double Differential Systematics (xF and Mass) - LH2 (5/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.15, 0.20)	[6.70, 6.80)	6.3086e-02	2.0925e-02	33.17%
[0.15, 0.20)	[7.00, 7.10)	3.7436e-02	1.1593e-02	30.97%
[0.15, 0.20)	[7.10, 7.20)	6.5899e-02	2.5087e-02	38.07%
[0.20, 0.25)	[4.30, 4.40)	1.0416e+00	4.7415e-01	45.52%
[0.20, 0.25)	[4.70, 4.80)	6.5499e-01	3.0186e-01	46.09%
[0.20, 0.25)	[4.80, 4.90)	4.2653e-01	1.3491e-01	31.63%
[0.20, 0.25)	[4.90, 5.00)	5.3867e-01	2.1491e-01	39.90%
[0.20, 0.25)	[5.00, 5.10)	4.8021e-01	2.0580e-01	42.86%
[0.20, 0.25)	[5.10, 5.20)	2.8186e-01	7.3912e-02	26.22%
[0.20, 0.25)	[5.20, 5.30)	3.8748e-01	7.0290e-02	18.14%
[0.20, 0.25)	[5.30, 5.40)	3.0340e-01	1.1571e-01	38.14%

Double Differential Systematics (xF and Mass) - LH2 (6/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.20, 0.25)	[5.50, 5.60)	2.6286e-01	3.5773e-02	13.61%
[0.20, 0.25)	[5.60, 5.70)	2.1079e-01	9.5591e-02	45.35%
[0.20, 0.25)	[5.70, 5.80)	1.5623e-01	7.0043e-02	44.83%
[0.20, 0.25)	[6.10, 6.20)	9.4111e-02	4.1035e-02	43.60%
[0.20, 0.25)	[6.20, 6.30)	1.1413e-01	4.6684e-02	40.91%
[0.20, 0.25)	[6.30, 6.40)	8.1846e-02	3.3641e-02	41.10%
[0.20, 0.25)	[6.70, 6.80)	2.4603e-02	5.3830e-03	21.88%
[0.20, 0.25)	[6.80, 6.90)	5.2887e-02	1.1228e-02	21.23%
[0.25, 0.30)	[4.20, 4.30)	3.8562e-01	1.8967e-01	49.19%
[0.25, 0.30)	[4.30, 4.40)	7.5327e-01	2.6101e-01	34.65%
[0.25, 0.30)	[4.40, 4.50)	1.1790e+00	3.1607e-01	26.81%

Double Differential Systematics (xF and Mass) - LH2 (7/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.25, 0.30)	[4.70, 4.80)	7.9801e-01	5.0829e-02	6.37%
[0.25, 0.30)	[4.80, 4.90)	4.2654e-01	9.0222e-02	21.15%
[0.25, 0.30)	[4.90, 5.00)	4.4999e-01	1.2228e-01	27.17%
[0.25, 0.30)	[5.00, 5.10)	4.8584e-01	1.2864e-01	26.48%
[0.25, 0.30)	[5.10, 5.20)	2.2713e-01	8.3800e-02	36.90%
[0.25, 0.30)	[5.20, 5.30)	2.4034e-01	9.5775e-02	39.85%
[0.25, 0.30)	[5.40, 5.50)	2.3389e-01	5.6802e-02	24.29%
[0.25, 0.30)	[5.60, 5.70)	2.4768e-01	9.1236e-02	36.84%
[0.25, 0.30)	[5.70, 5.80)	1.3372e-01	3.9861e-02	29.81%
[0.25, 0.30)	[5.80, 5.90)	1.6787e-01	2.7584e-02	16.43%
[0.25, 0.30)	[6.00, 6.10)	8.0099e-02	3.6185e-02	45.18%

Double Differential Systematics (xF and Mass) - LH2 (8/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.25, 0.30)	[6.20, 6.30)	8.3650e-02	3.8997e-02	46.62%
[0.25, 0.30)	[6.50, 6.60)	5.5701e-02	6.5372e-03	11.74%
[0.25, 0.30)	[6.70, 6.80)	3.5322e-02	1.1680e-02	33.07%
[0.25, 0.30)	[6.90, 7.00)	4.0496e-02	6.1293e-04	1.51%
[0.25, 0.30)	[7.00, 7.10)	4.7562e-02	2.3413e-02	49.23%
[0.25, 0.30)	[7.10, 7.20)	2.5905e-02	4.2366e-03	16.35%
[0.30, 0.35)	[4.20, 4.30)	7.3343e-01	1.9514e-01	26.61%
[0.30, 0.35)	[4.30, 4.40)	8.1888e-01	2.2517e-01	27.50%
[0.30, 0.35)	[4.40, 4.50)	8.5100e-01	3.9202e-01	46.07%
[0.30, 0.35)	[4.50, 4.60)	5.2640e-01	1.2486e-01	23.72%
[0.30, 0.35)	[4.60, 4.70)	5.6306e-01	1.1127e-01	19.76%

Double Differential Systematics (xF and Mass) - LH2 (9/24)

xF Range	Mass Range [GeV]	Mean σ_W	Std Dev $\sigma_{sys,w}$	Relative Error
[0.30, 0.35)	[4.80, 4.90)	4.3892e-01	1.0780e-01	24.56%
[0.30, 0.35)	[4.90, 5.00)	4.1002e-01	1.2520e-01	30.54%
[0.30, 0.35)	[5.00, 5.10)	4.0210e-01	1.4134e-01	35.15%
[0.30, 0.35)	[5.10, 5.20)	2.9704e-01	6.7877e-02	22.85%
[0.30, 0.35)	[5.20, 5.30)	3.0835e-01	6.2384e-02	20.23%
[0.30, 0.35)	[5.30, 5.40)	3.4379e-01	9.8521e-02	28.66%
[0.30, 0.35)	[5.40, 5.50)	1.9726e-01	4.0538e-02	20.55%
[0.30, 0.35)	[5.50, 5.60)	2.4710e-01	5.4274e-02	21.96%
[0.30, 0.35)	[5.60, 5.70)	1.6919e-01	3.7874e-02	22.39%
[0.30, 0.35)	[5.90, 6.00)	1.1824e-01	2.4555e-02	20.77%
[0.30, 0.35)	[6.00, 6.10)	1.0418e-01	3.1322e-02	30.07%

Double Differential Systematics (xF and Mass) - LH2 (10/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.30, 0.35)	[6.20, 6.30)	7.2372e-02	2.9229e-02	40.39%
[0.30, 0.35)	[6.30, 6.40)	9.4485e-02	3.3588e-02	35.55%
[0.30, 0.35)	[6.40, 6.50)	8.8967e-02	2.5721e-02	28.91%
[0.30, 0.35)	[6.50, 6.60)	7.5764e-02	2.3610e-02	31.16%
[0.30, 0.35)	[6.90, 7.00)	6.8826e-02	1.9560e-03	2.84%
[0.35, 0.40)	[4.20, 4.30)	6.6633e-01	1.3071e-01	19.62%
[0.35, 0.40)	[4.30, 4.40)	8.2753e-01	1.7939e-01	21.68%
[0.35, 0.40)	[4.40, 4.50)	6.0405e-01	1.7904e-01	29.64%
[0.35, 0.40)	[4.60, 4.70)	5.2074e-01	8.4980e-02	16.32%
[0.35, 0.40)	[4.70, 4.80)	5.4361e-01	5.5279e-02	10.17%
[0.35, 0.40)	[4.80, 4.90)	3.9604e-01	1.0977e-01	27.72%

Double Differential Systematics (xF and Mass) - LH2 (11/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.35, 0.40)	[5.00, 5.10)	3.8428e-01	1.1275e-01	29.34%
[0.35, 0.40)	[5.10, 5.20)	2.4910e-01	1.0098e-01	40.54%
[0.35, 0.40)	[5.20, 5.30)	2.6182e-01	9.5584e-02	36.51%
[0.35, 0.40)	[5.30, 5.40)	2.5192e-01	3.2237e-02	12.80%
[0.35, 0.40)	[5.40, 5.50)	1.8690e-01	4.3661e-02	23.36%
[0.35, 0.40)	[5.50, 5.60)	1.0287e-01	5.1401e-02	49.97%
[0.35, 0.40)	[5.60, 5.70)	1.1714e-01	5.0689e-02	43.27%
[0.35, 0.40)	[5.70, 5.80)	1.6693e-01	4.2988e-02	25.75%
[0.35, 0.40)	[5.80, 5.90)	9.7598e-02	4.8594e-02	49.79%
[0.35, 0.40)	[6.00, 6.10)	1.1606e-01	2.3183e-02	19.98%
[0.35, 0.40)	[6.10, 6.20)	8.4367e-02	2.1629e-02	25.64%

Double Differential Systematics (xF and Mass) - LH2 (12/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.35, 0.40)	[6.40, 6.50)	6.2507e-02	1.9807e-02	31.69%
[0.35, 0.40)	[6.60, 6.70)	4.8462e-02	1.4561e-02	30.05%
[0.35, 0.40)	[6.70, 6.80)	3.9186e-02	1.5855e-02	40.46%
[0.35, 0.40)	[7.00, 7.10)	2.6321e-02	9.8052e-05	0.37%
[0.35, 0.40)	[7.50, 7.60)	1.9093e-02	5.3499e-03	28.02%
[0.35, 0.40)	[7.60, 7.70)	2.5018e-02	9.3306e-05	0.37%
[0.40, 0.45)	[4.20, 4.30)	6.5020e-01	1.0341e-01	15.90%
[0.40, 0.45)	[4.30, 4.40)	6.4284e-01	7.3728e-02	11.47%
[0.40, 0.45)	[4.40, 4.50)	6.6360e-01	7.8016e-02	11.76%
[0.40, 0.45)	[4.50, 4.60)	5.2483e-01	1.1765e-01	22.42%
[0.40, 0.45)	[4.60, 4.70)	4.4001e-01	1.8037e-01	40.99%

Double Differential Systematics (xF and Mass) - LH2 (13/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.40, 0.45)	[4.90, 5.00)	2.7839e-01	4.8113e-02	17.28%
[0.40, 0.45)	[5.00, 5.10)	2.5497e-01	9.8386e-02	38.59%
[0.40, 0.45)	[5.10, 5.20)	2.3014e-01	8.1981e-02	35.62%
[0.40, 0.45)	[5.20, 5.30)	2.6026e-01	6.0031e-02	23.07%
[0.40, 0.45)	[5.30, 5.40)	2.0255e-01	5.1866e-02	25.61%
[0.40, 0.45)	[5.40, 5.50)	2.1960e-01	7.6062e-02	34.64%
[0.40, 0.45)	[5.50, 5.60)	1.6530e-01	7.9014e-02	47.80%
[0.40, 0.45)	[5.80, 5.90)	1.0860e-01	3.8835e-02	35.76%
[0.40, 0.45)	[5.90, 6.00)	7.8785e-02	3.0445e-02	38.64%
[0.40, 0.45)	[6.00, 6.10)	8.5339e-02	4.0094e-02	46.98%
[0.40, 0.45)	[6.30, 6.40)	1.0056e-01	3.3185e-02	33.00%

Double Differential Systematics (xF and Mass) - LH2 (14/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.40, 0.45)	[6.60, 6.70)	4.5108e-02	1.7759e-02	39.37%
[0.40, 0.45)	[6.80, 6.90)	3.3748e-02	1.5939e-02	47.23%
[0.40, 0.45)	[7.10, 7.20)	2.6293e-02	7.0993e-03	27.00%
[0.40, 0.45)	[7.20, 7.30)	3.7762e-02	1.7960e-02	47.56%
[0.45, 0.50)	[4.20, 4.30)	4.4186e-01	1.1267e-01	25.50%
[0.45, 0.50)	[4.30, 4.40)	5.3786e-01	1.1969e-01	22.25%
[0.45, 0.50)	[4.40, 4.50)	4.1364e-01	5.8842e-02	14.23%
[0.45, 0.50)	[4.50, 4.60)	4.7214e-01	1.1995e-01	25.40%
[0.45, 0.50)	[4.60, 4.70)	4.0952e-01	7.6236e-02	18.62%
[0.45, 0.50)	[4.70, 4.80)	3.4943e-01	1.6274e-01	46.57%
[0.45, 0.50)	[4.80, 4.90)	2.8876e-01	9.3421e-02	32.35%

Double Differential Systematics (xF and Mass) - LH2 (15/24)

xF Range	Mass Range [GeV]	Mean σ_W	Std Dev $\sigma_{sys,w}$	Relative Error
[0.45, 0.50)	[5.00, 5.10)	2.9148e-01	6.3792e-02	21.89%
[0.45, 0.50)	[5.10, 5.20)	1.8615e-01	3.9381e-02	21.16%
[0.45, 0.50)	[5.20, 5.30)	1.9278e-01	4.8492e-02	25.15%
[0.45, 0.50)	[5.30, 5.40)	1.7768e-01	1.6872e-02	9.50%
[0.45, 0.50)	[5.50, 5.60)	1.4331e-01	4.6619e-02	32.53%
[0.45, 0.50)	[5.60, 5.70)	1.0649e-01	2.7665e-02	25.98%
[0.45, 0.50)	[5.70, 5.80)	9.4103e-02	3.5888e-02	38.14%
[0.45, 0.50)	[5.90, 6.00)	7.7600e-02	3.3765e-02	43.51%
[0.45, 0.50)	[6.00, 6.10)	7.9333e-02	1.4629e-02	18.44%
[0.45, 0.50)	[6.20, 6.30)	5.2954e-02	1.7473e-02	33.00%
[0.45, 0.50)	[6.50, 6.60)	3.9426e-02	1.2829e-02	32.54%

Double Differential Systematics (xF and Mass) - LH2 (16/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.45, 0.50)	[7.20, 7.30)	3.2048e-02	1.0317e-02	32.19%
[0.45, 0.50)	[7.30, 7.40)	2.6372e-02	1.4723e-03	5.58%
[0.50, 0.55)	[4.20, 4.30)	4.7043e-01	1.0073e-01	21.41%
[0.50, 0.55)	[4.30, 4.40)	5.4145e-01	9.5272e-02	17.60%
[0.50, 0.55)	[4.40, 4.50)	5.3161e-01	8.3148e-02	15.64%
[0.50, 0.55)	[4.50, 4.60)	2.4280e-01	9.3307e-02	38.43%
[0.50, 0.55)	[4.60, 4.70)	2.3730e-01	3.6697e-02	15.46%
[0.50, 0.55)	[4.80, 4.90)	2.0232e-01	4.6960e-02	23.21%
[0.50, 0.55)	[4.90, 5.00)	2.2458e-01	4.5678e-02	20.34%
[0.50, 0.55)	[5.00, 5.10)	1.8493e-01	6.1855e-02	33.45%
[0.50, 0.55)	[5.10, 5.20)	1.8434e-01	6.9482e-02	37.69%

Double Differential Systematics (xF and Mass) - LH2 (17/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.50, 0.55)	[5.30, 5.40)	1.7399e-01	6.7936e-02	39.05%
[0.50, 0.55)	[5.40, 5.50)	1.0882e-01	3.9254e-02	36.07%
[0.50, 0.55)	[5.50, 5.60)	1.0260e-01	3.9398e-02	38.40%
[0.50, 0.55)	[5.60, 5.70)	8.2846e-02	2.0093e-02	24.25%
[0.50, 0.55)	[5.70, 5.80)	7.9674e-02	2.3015e-02	28.89%
[0.50, 0.55)	[5.80, 5.90)	4.8213e-02	8.2614e-03	17.14%
[0.50, 0.55)	[5.90, 6.00)	6.2108e-02	1.7050e-02	27.45%
[0.50, 0.55)	[6.00, 6.10)	5.7448e-02	6.0101e-03	10.46%
[0.50, 0.55)	[6.10, 6.20)	4.7435e-02	1.5387e-02	32.44%
[0.50, 0.55)	[6.40, 6.50)	3.4890e-02	7.7363e-03	22.17%
[0.50, 0.55)	[6.50, 6.60)	3.8857e-02	1.8553e-02	47.75%

Double Differential Systematics (xF and Mass) - LH2 (18/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.50, 0.55)	[7.20, 7.30)	1.7977e-02	7.0583e-03	39.26%
[0.55, 0.60)	[4.20, 4.30)	3.1446e-01	2.8356e-02	9.02%
[0.55, 0.60)	[4.30, 4.40)	3.4566e-01	7.2500e-02	20.97%
[0.55, 0.60)	[4.40, 4.50)	3.1599e-01	7.8161e-02	24.74%
[0.55, 0.60)	[4.60, 4.70)	2.4332e-01	4.4295e-02	18.20%
[0.55, 0.60)	[4.70, 4.80)	2.0502e-01	5.6428e-02	27.52%
[0.55, 0.60)	[4.80, 4.90)	1.5930e-01	4.5898e-02	28.81%
[0.55, 0.60)	[4.90, 5.00)	1.4238e-01	2.8971e-02	20.35%
[0.55, 0.60)	[5.00, 5.10)	1.9249e-01	3.6309e-02	18.86%
[0.55, 0.60)	[5.10, 5.20)	1.3331e-01	3.7765e-02	28.33%
[0.55, 0.60)	[5.20, 5.30)	1.1815e-01	4.5417e-02	38.44%

Double Differential Systematics (xF and Mass) - LH2 (19/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.55, 0.60)	[5.40, 5.50)	7.9734e-02	2.1307e-02	26.72%
[0.55, 0.60)	[5.50, 5.60)	6.6886e-02	2.7636e-02	41.32%
[0.55, 0.60)	[5.70, 5.80)	4.1620e-02	1.3927e-02	33.46%
[0.55, 0.60)	[5.90, 6.00)	4.7955e-02	1.8641e-02	38.87%
[0.55, 0.60)	[6.10, 6.20)	3.1484e-02	1.1781e-02	37.42%
[0.55, 0.60)	[6.20, 6.30)	2.9954e-02	1.4713e-02	49.12%
[0.55, 0.60)	[6.50, 6.60)	2.2299e-02	7.7467e-03	34.74%
[0.55, 0.60)	[6.60, 6.70)	4.7309e-02	2.2805e-02	48.20%
[0.55, 0.60)	[7.50, 7.60)	2.2412e-02	3.0111e-03	13.44%
[0.60, 0.65)	[4.20, 4.30)	2.3237e-01	6.5012e-02	27.98%
[0.60, 0.65)	[4.40, 4.50)	2.8412e-01	5.0776e-02	17.87%

Double Differential Systematics (xF and Mass) - LH2 (20/24)

xF Range	Mass Range [GeV]	Mean σ_W	Std Dev $\sigma_{sys,w}$	Relative Error
[0.60, 0.65)	[4.60, 4.70)	1.6645e-01	5.2879e-02	31.77%
[0.60, 0.65)	[4.70, 4.80)	1.9743e-01	2.4467e-02	12.39%
[0.60, 0.65)	[4.80, 4.90)	9.7683e-02	3.9021e-02	39.95%
[0.60, 0.65)	[4.90, 5.00)	1.1797e-01	4.6537e-02	39.45%
[0.60, 0.65)	[5.00, 5.10)	1.0068e-01	1.4906e-02	14.81%
[0.60, 0.65)	[5.10, 5.20)	7.9764e-02	2.6604e-02	33.35%
[0.60, 0.65)	[5.20, 5.30)	8.2359e-02	1.8714e-02	22.72%
[0.60, 0.65)	[5.30, 5.40)	6.9428e-02	3.1077e-02	44.76%
[0.60, 0.65)	[5.40, 5.50)	6.7480e-02	2.1666e-02	32.11%
[0.60, 0.65)	[5.50, 5.60)	5.1558e-02	2.3491e-02	45.56%
[0.60, 0.65)	[5.60, 5.70)	4.7267e-02	1.4393e-02	30.45%

Double Differential Systematics (xF and Mass) - LH2 (21/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.60, 0.65)	[6.30, 6.40)	3.0712e-02	1.4422e-02	46.96%
[0.60, 0.65)	[7.10, 7.20)	1.4250e-02	8.9948e-04	6.31%
[0.65, 0.70)	[4.20, 4.30)	1.6306e-01	3.8978e-02	23.90%
[0.65, 0.70)	[4.30, 4.40)	1.5587e-01	9.9566e-03	6.39%
[0.65, 0.70)	[4.40, 4.50)	1.3019e-01	3.6288e-02	27.87%
[0.65, 0.70)	[4.50, 4.60)	1.1632e-01	1.5350e-02	13.20%
[0.65, 0.70)	[4.60, 4.70)	1.1617e-01	5.1048e-02	43.94%
[0.65, 0.70)	[4.70, 4.80)	1.2258e-01	2.3437e-02	19.12%
[0.65, 0.70)	[4.80, 4.90)	9.5394e-02	2.0905e-02	21.91%
[0.65, 0.70)	[4.90, 5.00)	4.0291e-02	1.9543e-02	48.51%
[0.65, 0.70)	[5.00, 5.10)	6.3671e-02	3.0105e-02	47.28%

Double Differential Systematics (xF and Mass) - LH2 (22/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.65, 0.70)	[5.30, 5.40)	4.2494e-02	1.4566e-02	34.28%
[0.65, 0.70)	[5.40, 5.50)	5.1017e-02	1.9123e-02	37.48%
[0.65, 0.70)	[5.60, 5.70)	4.8786e-02	1.4792e-02	30.32%
[0.65, 0.70)	[5.70, 5.80)	3.4528e-02	9.2034e-03	26.65%
[0.65, 0.70)	[5.90, 6.00)	2.3854e-02	1.1355e-02	47.60%
[0.65, 0.70)	[6.00, 6.10)	2.0751e-02	5.5300e-03	26.65%
[0.65, 0.70)	[6.10, 6.20)	2.4947e-02	6.3603e-03	25.50%
[0.65, 0.70)	[6.70, 6.80)	1.8812e-02	8.3791e-03	44.54%
[0.70, 0.75)	[4.20, 4.30)	1.1624e-01	2.4962e-02	21.47%
[0.70, 0.75)	[4.30, 4.40)	1.2383e-01	1.5991e-02	12.91%
[0.70, 0.75)	[4.40, 4.50)	8.3088e-02	3.7542e-02	45.18%

Double Differential Systematics (xF and Mass) - LH2 (23/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.70, 0.75)	[4.60, 4.70)	5.9294e-02	2.0304e-02	34.24%
[0.70, 0.75)	[4.70, 4.80)	6.3518e-02	2.7601e-02	43.45%
[0.70, 0.75)	[4.80, 4.90)	6.3039e-02	2.0785e-02	32.97%
[0.70, 0.75)	[4.90, 5.00)	4.6842e-02	9.3913e-03	20.05%
[0.70, 0.75)	[5.20, 5.30)	1.8832e-02	7.1401e-03	37.92%
[0.70, 0.75)	[5.50, 5.60)	1.9884e-02	8.4498e-03	42.50%
[0.70, 0.75)	[5.60, 5.70)	3.1179e-02	1.3155e-02	42.19%
[0.70, 0.75)	[5.90, 6.00)	1.6568e-02	6.0635e-03	36.60%
[0.70, 0.75)	[6.00, 6.10)	1.7038e-02	6.7872e-03	39.84%
[0.70, 0.75)	[6.10, 6.20)	2.4911e-02	5.6151e-03	22.54%
[0.70, 0.75)	[6.40, 6.50)	2.0956e-02	9.1818e-03	43.82%

Double Differential Systematics (xF and Mass) - LH2 (24/24)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.75, 0.80)	[4.20, 4.30)	6.7594e-02	1.7019e-02	25.18%
[0.75, 0.80)	[4.30, 4.40)	4.7527e-02	1.9481e-02	40.99%
[0.75, 0.80)	[4.60, 4.70)	2.1578e-02	1.0617e-02	49.20%
[0.75, 0.80)	[4.90, 5.00)	2.7924e-02	5.7338e-03	20.53%
[0.75, 0.80)	[5.30, 5.40)	2.0334e-02	8.4083e-03	41.35%
[0.75, 0.80)	[5.40, 5.50)	2.2852e-02	8.4208e-03	36.85%
[0.75, 0.80)	[5.50, 5.60)	1.4216e-02	3.8382e-03	27.00%
[0.75, 0.80)	[5.70, 5.80)	1.8674e-02	6.8121e-03	36.48%
[0.75, 0.80)	[5.90, 6.00)	1.3990e-02	5.1247e-03	36.63%

Double Differential Systematics (xF and Mass) - LD2 (1/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.00, 0.05)	[5.00, 5.10)	1.8116e+00	2.9436e-01	16.25%
[0.00, 0.05)	[5.30, 5.40)	7.4077e-01	2.5270e-01	34.11%
[0.00, 0.05)	[5.70, 5.80)	2.7498e-01	7.9064e-02	28.75%
[0.00, 0.05)	[6.00, 6.10)	2.4899e-01	1.1419e-01	45.86%
[0.00, 0.05)	[6.10, 6.20)	2.9414e-01	4.6021e-02	15.65%
[0.00, 0.05)	[6.40, 6.50)	2.0379e-01	2.6151e-02	12.83%
[0.00, 0.05)	[6.50, 6.60)	2.1517e-01	6.5513e-02	30.45%
[0.00, 0.05)	[7.10, 7.20)	1.4900e-01	1.5269e-02	10.25%
[0.05, 0.10)	[4.40, 4.50)	1.2721e+01	5.9172e+00	46.52%
[0.05, 0.10)	[4.80, 4.90)	1.3710e+00	3.2997e-01	24.07%
[0.05, 0.10)	[4.90, 5.00)	1.4715e+00	5.3111e-01	36.09%

Double Differential Systematics (xF and Mass) - LD2 (2/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.05, 0.10)	[5.10, 5.20)	5.9756e-01	7.5644e-02	12.66%
[0.05, 0.10)	[5.20, 5.30)	5.8076e-01	2.4453e-01	42.10%
[0.05, 0.10)	[5.50, 5.60)	3.6514e-01	1.6532e-01	45.28%
[0.05, 0.10)	[6.00, 6.10)	2.1293e-01	4.3579e-02	20.47%
[0.05, 0.10)	[6.10, 6.20)	2.2400e-01	1.0773e-01	48.09%
[0.05, 0.10)	[6.30, 6.40)	3.0125e-01	9.3422e-02	31.01%
[0.05, 0.10)	[6.40, 6.50)	1.8374e-01	6.2461e-02	33.99%
[0.05, 0.10)	[6.70, 6.80)	1.4736e-01	5.6844e-02	38.57%
[0.05, 0.10)	[6.80, 6.90)	9.1846e-02	2.9942e-02	32.60%
[0.10, 0.15)	[4.20, 4.30)	1.5886e+00	3.9309e-01	24.74%
[0.10, 0.15)	[4.50, 4.60)	1.1752e+00	4.4030e-01	37.47%

Double Differential Systematics (xF and Mass) - LD2 (3/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.10, 0.15)	[4.80, 4.90)	9.1156e-01	3.1401e-01	34.45%
[0.10, 0.15)	[4.90, 5.00)	1.2460e+00	4.2808e-01	34.36%
[0.10, 0.15)	[5.00, 5.10)	1.7892e+00	5.8412e-01	32.65%
[0.10, 0.15)	[5.10, 5.20)	8.7170e-01	2.6662e-01	30.59%
[0.10, 0.15)	[5.20, 5.30)	1.0346e+00	1.7277e-01	16.70%
[0.10, 0.15)	[5.30, 5.40)	7.1808e-01	2.7124e-01	37.77%
[0.10, 0.15)	[5.40, 5.50)	4.7704e-01	9.5059e-02	19.93%
[0.10, 0.15)	[5.50, 5.60)	4.5765e-01	1.2112e-01	26.47%
[0.10, 0.15)	[5.60, 5.70)	3.6645e-01	7.0209e-02	19.16%
[0.10, 0.15)	[5.70, 5.80)	3.4852e-01	1.4016e-01	40.22%
[0.10, 0.15)	[5.80, 5.90)	3.0950e-01	6.3485e-02	20.51%

Double Differential Systematics (xF and Mass) - LD2 (4/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.10, 0.15)	[6.10, 6.20)	2.5534e-01	9.0534e-02	35.46%
[0.10, 0.15)	[6.20, 6.30)	2.3110e-01	6.3075e-02	27.29%
[0.10, 0.15)	[6.70, 6.80)	1.9230e-01	8.4116e-02	43.74%
[0.10, 0.15)	[6.80, 6.90)	1.6123e-01	6.6851e-02	41.46%
[0.10, 0.15)	[6.90, 7.00)	1.2891e-01	5.3148e-02	41.23%
[0.15, 0.20)	[4.40, 4.50)	1.7209e+00	5.2334e-01	30.41%
[0.15, 0.20)	[4.60, 4.70)	2.0494e+00	5.0075e-01	24.43%
[0.15, 0.20)	[4.70, 4.80)	1.7576e+00	7.6632e-01	43.60%
[0.15, 0.20)	[4.80, 4.90)	1.1816e+00	4.1852e-01	35.42%
[0.15, 0.20)	[4.90, 5.00)	1.2338e+00	2.9127e-01	23.61%
[0.15, 0.20)	[5.00, 5.10)	1.6994e+00	3.8863e-01	22.87%

Double Differential Systematics (xF and Mass) - LD2 (5/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.15, 0.20)	[5.30, 5.40)	9.3847e-01	1.6866e-01	17.97%
[0.15, 0.20)	[5.40, 5.50)	3.9759e-01	1.0897e-01	27.41%
[0.15, 0.20)	[5.50, 5.60)	4.2744e-01	1.9178e-01	44.87%
[0.15, 0.20)	[5.60, 5.70)	5.2308e-01	1.1058e-01	21.14%
[0.15, 0.20)	[5.70, 5.80)	3.3979e-01	1.5443e-01	45.45%
[0.15, 0.20)	[5.90, 6.00)	2.7808e-01	8.9340e-02	32.13%
[0.15, 0.20)	[6.00, 6.10)	2.6297e-01	6.0843e-02	23.14%
[0.15, 0.20)	[6.40, 6.50)	2.4188e-01	8.6470e-02	35.75%
[0.15, 0.20)	[6.60, 6.70)	6.7705e-02	2.8814e-02	42.56%
[0.15, 0.20)	[6.70, 6.80)	1.8154e-01	4.0359e-02	22.23%
[0.15, 0.20)	[6.80, 6.90)	8.4475e-02	2.2659e-02	26.82%

Double Differential Systematics (xF and Mass) - LD2 (6/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.15, 0.20)	[7.20, 7.30)	6.3309e-02	1.6316e-03	2.58%
[0.15, 0.20)	[7.60, 7.70)	1.0333e-01	2.9103e-02	28.17%
[0.15, 0.20)	[7.70, 7.80)	1.3385e-01	2.6271e-02	19.63%
[0.20, 0.25)	[4.20, 4.30)	1.9662e+00	4.7737e-01	24.28%
[0.20, 0.25)	[4.40, 4.50)	2.7188e+00	9.4080e-01	34.60%
[0.20, 0.25)	[4.50, 4.60)	1.2256e+00	5.1222e-01	41.79%
[0.20, 0.25)	[4.60, 4.70)	1.2279e+00	2.3360e-01	19.02%
[0.20, 0.25)	[4.70, 4.80)	1.7695e+00	4.0219e-01	22.73%
[0.20, 0.25)	[4.80, 4.90)	1.0702e+00	5.3039e-01	49.56%
[0.20, 0.25)	[4.90, 5.00)	1.1543e+00	3.4702e-01	30.06%
[0.20, 0.25)	[5.00, 5.10)	1.0012e+00	3.5544e-01	35.50%

Double Differential Systematics (xF and Mass) - LD2 (7/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.20, 0.25)	[5.20, 5.30)	8.1372e-01	3.1610e-01	38.85%
[0.20, 0.25)	[5.30, 5.40)	6.8434e-01	1.5386e-01	22.48%
[0.20, 0.25)	[5.50, 5.60)	5.6949e-01	9.7198e-02	17.07%
[0.20, 0.25)	[5.60, 5.70)	4.5872e-01	1.0129e-01	22.08%
[0.20, 0.25)	[5.70, 5.80)	4.7841e-01	1.1290e-01	23.60%
[0.20, 0.25)	[5.80, 5.90)	2.8679e-01	1.1389e-01	39.71%
[0.20, 0.25)	[6.00, 6.10)	3.4276e-01	6.2581e-02	18.26%
[0.20, 0.25)	[6.10, 6.20)	2.8220e-01	6.5846e-02	23.33%
[0.20, 0.25)	[6.20, 6.30)	1.2312e-01	5.4644e-02	44.38%
[0.20, 0.25)	[6.30, 6.40)	2.8960e-01	1.2414e-01	42.87%
[0.20, 0.25)	[6.50, 6.60)	1.4064e-01	5.2658e-02	37.44%

Double Differential Systematics (xF and Mass) - LD2 (8/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.20, 0.25)	[6.80, 6.90)	7.3027e-02	2.3730e-02	32.50%
[0.20, 0.25)	[6.90, 7.00)	1.1827e-01	5.6980e-02	48.18%
[0.20, 0.25)	[7.10, 7.20)	6.1944e-02	2.5528e-02	41.21%
[0.20, 0.25)	[7.40, 7.50)	5.3967e-02	2.2095e-02	40.94%
[0.25, 0.30)	[4.20, 4.30)	2.1312e+00	7.0669e-01	33.16%
[0.25, 0.30)	[4.30, 4.40)	2.3751e+00	6.6329e-01	27.93%
[0.25, 0.30)	[4.40, 4.50)	2.6323e+00	8.9668e-01	34.06%
[0.25, 0.30)	[4.50, 4.60)	1.4308e+00	3.1528e-01	22.04%
[0.25, 0.30)	[4.60, 4.70)	1.2488e+00	3.5644e-01	28.54%
[0.25, 0.30)	[4.70, 4.80)	1.5853e+00	4.5268e-01	28.56%
[0.25, 0.30)	[4.80, 4.90)	8.5633e-01	3.3308e-01	38.90%

Double Differential Systematics (xF and Mass) - LD2 (9/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.25, 0.30)	[5.00, 5.10)	9.8256e-01	1.5859e-01	16.14%
[0.25, 0.30)	[5.10, 5.20)	6.2821e-01	1.2756e-01	20.31%
[0.25, 0.30)	[5.20, 5.30)	6.7952e-01	2.5413e-01	37.40%
[0.25, 0.30)	[5.30, 5.40)	5.1098e-01	1.1191e-01	21.90%
[0.25, 0.30)	[5.40, 5.50)	4.7269e-01	1.3052e-01	27.61%
[0.25, 0.30)	[5.50, 5.60)	3.5863e-01	1.6390e-01	45.70%
[0.25, 0.30)	[5.60, 5.70)	4.0188e-01	8.1073e-02	20.17%
[0.25, 0.30)	[5.70, 5.80)	4.0316e-01	4.5732e-02	11.34%
[0.25, 0.30)	[6.00, 6.10)	2.6213e-01	2.6205e-02	10.00%
[0.25, 0.30)	[6.10, 6.20)	2.9041e-01	1.1796e-01	40.62%
[0.25, 0.30)	[6.20, 6.30)	2.2724e-01	2.0976e-02	9.23%

Double Differential Systematics (xF and Mass) - LD2 (10/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.25, 0.30)	[6.70, 6.80)	9.2486e-02	2.0953e-02	22.66%
[0.25, 0.30)	[6.90, 7.00)	9.3549e-02	3.3863e-02	36.20%
[0.25, 0.30)	[7.00, 7.10)	9.2873e-02	2.5101e-02	27.03%
[0.25, 0.30)	[7.20, 7.30)	7.7751e-02	8.3349e-03	10.72%
[0.25, 0.30)	[7.30, 7.40)	8.0221e-02	1.5441e-02	19.25%
[0.30, 0.35)	[4.30, 4.40)	1.8637e+00	2.7665e-01	14.84%
[0.30, 0.35)	[4.40, 4.50)	1.9054e+00	3.1579e-01	16.57%
[0.30, 0.35)	[4.50, 4.60)	1.2027e+00	2.0372e-01	16.94%
[0.30, 0.35)	[4.60, 4.70)	1.0523e+00	3.4276e-01	32.57%
[0.30, 0.35)	[4.70, 4.80)	1.2913e+00	5.9478e-01	46.06%
[0.30, 0.35)	[4.80, 4.90)	1.0451e+00	3.6769e-01	35.18%

Double Differential Systematics (xF and Mass) - LD2 (11/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.30, 0.35)	[5.00, 5.10)	8.6194e-01	2.0553e-01	23.85%
[0.30, 0.35)	[5.20, 5.30)	6.0505e-01	1.4140e-01	23.37%
[0.30, 0.35)	[5.30, 5.40)	6.5364e-01	1.1992e-01	18.35%
[0.30, 0.35)	[5.50, 5.60)	4.0817e-01	1.0608e-01	25.99%
[0.30, 0.35)	[5.60, 5.70)	4.3239e-01	5.8004e-02	13.41%
[0.30, 0.35)	[5.70, 5.80)	2.9054e-01	1.2921e-01	44.47%
[0.30, 0.35)	[6.00, 6.10)	2.5977e-01	5.4340e-02	20.92%
[0.30, 0.35)	[6.10, 6.20)	2.5160e-01	8.8153e-02	35.04%
[0.30, 0.35)	[6.20, 6.30)	1.1839e-01	2.8686e-02	24.23%
[0.30, 0.35)	[6.30, 6.40)	2.1467e-01	8.2771e-02	38.56%
[0.30, 0.35)	[6.50, 6.60)	1.4160e-01	3.1720e-02	22.40%

Double Differential Systematics (xF and Mass) - LD2 (12/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.30, 0.35)	[7.00, 7.10)	6.4592e-02	2.1208e-02	32.83%
[0.30, 0.35)	[7.20, 7.30)	7.8270e-02	3.3428e-02	42.71%
[0.30, 0.35)	[7.50, 7.60)	6.5183e-02	9.5982e-03	14.72%
[0.30, 0.35)	[7.60, 7.70)	6.6621e-02	6.8519e-03	10.28%
[0.35, 0.40)	[4.20, 4.30)	1.5873e+00	2.4120e-01	15.20%
[0.35, 0.40)	[4.30, 4.40)	1.8208e+00	2.3248e-01	12.77%
[0.35, 0.40)	[4.40, 4.50)	1.7423e+00	5.0254e-01	28.84%
[0.35, 0.40)	[4.50, 4.60)	1.2608e+00	2.1801e-01	17.29%
[0.35, 0.40)	[4.60, 4.70)	1.2549e+00	2.5651e-01	20.44%
[0.35, 0.40)	[4.70, 4.80)	1.1491e+00	2.2613e-01	19.68%
[0.35, 0.40)	[4.80, 4.90)	9.2427e-01	1.4555e-01	15.75%

Double Differential Systematics (xF and Mass) - LD2 (13/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.35, 0.40)	[5.20, 5.30)	6.1219e-01	1.1665e-01	19.05%
[0.35, 0.40)	[5.40, 5.50)	3.9775e-01	1.7142e-01	43.10%
[0.35, 0.40)	[5.50, 5.60)	3.7351e-01	9.5437e-02	25.55%
[0.35, 0.40)	[5.70, 5.80)	3.0016e-01	1.2349e-01	41.14%
[0.35, 0.40)	[5.80, 5.90)	3.0814e-01	3.7375e-02	12.13%
[0.35, 0.40)	[5.90, 6.00)	2.5015e-01	1.1468e-01	45.84%
[0.35, 0.40)	[6.10, 6.20)	1.4764e-01	6.1723e-02	41.81%
[0.35, 0.40)	[6.20, 6.30)	1.3482e-01	4.5423e-02	33.69%
[0.35, 0.40)	[6.40, 6.50)	1.4286e-01	2.4717e-02	17.30%
[0.35, 0.40)	[6.60, 6.70)	1.2197e-01	4.6880e-02	38.44%
[0.35, 0.40)	[6.70, 6.80)	9.6300e-02	7.5793e-03	7.87%

Double Differential Systematics (xF and Mass) - LD2 (14/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.35, 0.40)	[6.90, 7.00)	4.9290e-02	2.9353e-03	5.96%
[0.35, 0.40)	[7.00, 7.10)	6.2159e-02	2.6764e-02	43.06%
[0.35, 0.40)	[7.10, 7.20)	8.8577e-02	2.4683e-02	27.87%
[0.35, 0.40)	[7.60, 7.70)	7.9243e-02	1.3427e-02	16.94%
[0.40, 0.45)	[4.20, 4.30)	1.5789e+00	2.0121e-01	12.74%
[0.40, 0.45)	[4.30, 4.40)	1.3675e+00	1.5673e-01	11.46%
[0.40, 0.45)	[4.40, 4.50)	1.3825e+00	2.8435e-01	20.57%
[0.40, 0.45)	[4.50, 4.60)	1.1425e+00	1.3999e-01	12.25%
[0.40, 0.45)	[4.60, 4.70)	1.2314e+00	6.0255e-02	4.89%
[0.40, 0.45)	[4.70, 4.80)	1.0438e+00	1.2191e-01	11.68%
[0.40, 0.45)	[4.80, 4.90)	8.4358e-01	2.2966e-01	27.22%

Double Differential Systematics (xF and Mass) - LD2 (15/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.40, 0.45)	[5.00, 5.10)	6.4615e-01	1.3114e-01	20.30%
[0.40, 0.45)	[5.10, 5.20)	5.1830e-01	8.7093e-02	16.80%
[0.40, 0.45)	[5.20, 5.30)	4.5336e-01	1.5200e-01	33.53%
[0.40, 0.45)	[5.30, 5.40)	5.6706e-01	2.4555e-01	43.30%
[0.40, 0.45)	[5.40, 5.50)	4.4768e-01	8.9087e-02	19.90%
[0.40, 0.45)	[5.50, 5.60)	3.5416e-01	1.1217e-01	31.67%
[0.40, 0.45)	[5.60, 5.70)	2.7690e-01	5.8492e-02	21.12%
[0.40, 0.45)	[5.80, 5.90)	2.2938e-01	5.1352e-02	22.39%
[0.40, 0.45)	[5.90, 6.00)	2.8875e-01	5.8819e-02	20.37%
[0.40, 0.45)	[6.10, 6.20)	1.6415e-01	6.9650e-02	42.43%
[0.40, 0.45)	[6.20, 6.30)	1.8752e-01	2.7736e-02	14.79%

Double Differential Systematics (xF and Mass) - LD2 (16/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.40, 0.45)	[6.60, 6.70)	1.0857e-01	1.4610e-02	13.46%
[0.40, 0.45)	[6.70, 6.80)	1.2123e-01	2.9287e-02	24.16%
[0.45, 0.50)	[4.20, 4.30)	1.1350e+00	1.9105e-01	16.83%
[0.45, 0.50)	[4.30, 4.40)	1.2826e+00	2.5349e-01	19.76%
[0.45, 0.50)	[4.40, 4.50)	1.1348e+00	1.5590e-01	13.74%
[0.45, 0.50)	[4.50, 4.60)	8.2839e-01	1.3537e-01	16.34%
[0.45, 0.50)	[4.60, 4.70)	8.4093e-01	2.5701e-01	30.56%
[0.45, 0.50)	[4.70, 4.80)	8.2277e-01	3.1013e-01	37.69%
[0.45, 0.50)	[4.80, 4.90)	5.8365e-01	1.5827e-01	27.12%
[0.45, 0.50)	[4.90, 5.00)	6.5023e-01	1.1931e-01	18.35%
[0.45, 0.50)	[5.00, 5.10)	6.4121e-01	1.8035e-01	28.13%

Double Differential Systematics (xF and Mass) - LD2 (17/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.45, 0.50)	[5.20, 5.30)	4.0113e-01	4.0472e-02	10.09%
[0.45, 0.50)	[5.30, 5.40)	4.0098e-01	1.4519e-01	36.21%
[0.45, 0.50)	[5.40, 5.50)	3.5241e-01	6.0967e-02	17.30%
[0.45, 0.50)	[5.50, 5.60)	2.8752e-01	8.4579e-02	29.42%
[0.45, 0.50)	[5.60, 5.70)	2.6489e-01	6.4035e-02	24.17%
[0.45, 0.50)	[5.80, 5.90)	2.5039e-01	5.7624e-02	23.01%
[0.45, 0.50)	[5.90, 6.00)	1.6974e-01	7.1841e-02	42.32%
[0.45, 0.50)	[6.10, 6.20)	1.5282e-01	2.4516e-02	16.04%
[0.45, 0.50)	[6.20, 6.30)	1.4595e-01	4.2283e-02	28.97%
[0.45, 0.50)	[6.40, 6.50)	1.2186e-01	3.7816e-02	31.03%
[0.45, 0.50)	[6.70, 6.80)	5.8257e-02	1.4236e-02	24.44%

Double Differential Systematics (xF and Mass) - LD2 (18/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.45, 0.50)	[7.00, 7.10)	6.6903e-02	2.0330e-02	30.39%
[0.45, 0.50)	[7.10, 7.20)	6.5860e-02	1.6052e-02	24.37%
[0.45, 0.50)	[7.30, 7.40)	5.3005e-02	1.8350e-02	34.62%
[0.50, 0.55)	[4.20, 4.30)	1.0880e+00	1.3435e-01	12.35%
[0.50, 0.55)	[4.30, 4.40)	9.8801e-01	2.7745e-01	28.08%
[0.50, 0.55)	[4.40, 4.50)	1.0079e+00	1.6536e-01	16.41%
[0.50, 0.55)	[4.50, 4.60)	7.2623e-01	9.0976e-02	12.53%
[0.50, 0.55)	[4.60, 4.70)	6.8292e-01	1.2543e-01	18.37%
[0.50, 0.55)	[4.70, 4.80)	7.6598e-01	1.5935e-01	20.80%
[0.50, 0.55)	[4.80, 4.90)	4.9332e-01	2.1142e-01	42.86%
[0.50, 0.55)	[4.90, 5.00)	4.5948e-01	2.2904e-01	49.85%

Double Differential Systematics (xF and Mass) - LD2 (19/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.50, 0.55)	[5.20, 5.30)	3.4870e-01	5.1268e-02	14.70%
[0.50, 0.55)	[5.30, 5.40)	3.6272e-01	8.0334e-02	22.15%
[0.50, 0.55)	[5.40, 5.50)	2.5445e-01	1.0288e-01	40.43%
[0.50, 0.55)	[5.50, 5.60)	1.9698e-01	8.2507e-02	41.88%
[0.50, 0.55)	[5.60, 5.70)	2.1040e-01	5.1877e-02	24.66%
[0.50, 0.55)	[5.70, 5.80)	1.4585e-01	7.1704e-02	49.16%
[0.50, 0.55)	[5.80, 5.90)	1.4911e-01	6.1392e-02	41.17%
[0.50, 0.55)	[5.90, 6.00)	1.9120e-01	6.5785e-02	34.41%
[0.50, 0.55)	[6.00, 6.10)	1.3142e-01	4.2475e-02	32.32%
[0.50, 0.55)	[6.10, 6.20)	1.5405e-01	3.6758e-02	23.86%
[0.50, 0.55)	[6.30, 6.40)	9.8446e-02	4.2197e-02	42.86%

Double Differential Systematics (xF and Mass) - LD2 (20/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.50, 0.55)	[6.60, 6.70)	1.0812e-01	1.8378e-02	17.00%
[0.50, 0.55)	[6.70, 6.80)	7.0812e-02	1.3028e-02	18.40%
[0.50, 0.55)	[6.80, 6.90)	7.9242e-02	3.1092e-02	39.24%
[0.50, 0.55)	[6.90, 7.00)	7.1701e-02	3.4834e-02	48.58%
[0.50, 0.55)	[7.80, 7.90)	3.7468e-02	9.2689e-03	24.74%
[0.55, 0.60)	[4.20, 4.30)	7.9794e-01	1.0174e-01	12.75%
[0.55, 0.60)	[4.30, 4.40)	6.3255e-01	1.7720e-01	28.01%
[0.55, 0.60)	[4.40, 4.50)	8.1346e-01	1.0594e-01	13.02%
[0.55, 0.60)	[4.50, 4.60)	5.3201e-01	1.3075e-01	24.58%
[0.55, 0.60)	[4.60, 4.70)	5.7427e-01	9.6132e-02	16.74%
[0.55, 0.60)	[4.70, 4.80)	3.9912e-01	7.3688e-02	18.46%

Double Differential Systematics (xF and Mass) - LD2 (21/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.55, 0.60)	[4.90, 5.00)	4.4323e-01	1.2363e-01	27.89%
[0.55, 0.60)	[5.00, 5.10)	4.8502e-01	9.4404e-02	19.46%
[0.55, 0.60)	[5.10, 5.20)	2.9059e-01	7.7698e-02	26.74%
[0.55, 0.60)	[5.20, 5.30)	2.5372e-01	7.0417e-02	27.75%
[0.55, 0.60)	[5.30, 5.40)	2.3984e-01	5.7677e-02	24.05%
[0.55, 0.60)	[5.40, 5.50)	2.2865e-01	4.1814e-02	18.29%
[0.55, 0.60)	[5.50, 5.60)	2.1740e-01	6.3168e-02	29.06%
[0.55, 0.60)	[5.70, 5.80)	1.0225e-01	1.9396e-02	18.97%
[0.55, 0.60)	[5.90, 6.00)	1.3543e-01	4.8516e-02	35.82%
[0.55, 0.60)	[6.00, 6.10)	1.1013e-01	3.3920e-02	30.80%
[0.55, 0.60)	[6.10, 6.20)	7.7386e-02	2.1430e-02	27.69%

Double Differential Systematics (xF and Mass) - LD2 (22/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.55, 0.60)	[6.60, 6.70)	7.1667e-02	1.6986e-02	23.70%
[0.55, 0.60)	[6.80, 6.90)	3.2194e-02	9.4070e-03	29.22%
[0.55, 0.60)	[7.10, 7.20)	6.2236e-02	3.1439e-03	5.05%
[0.55, 0.60)	[7.40, 7.50)	2.3738e-02	1.0202e-02	42.98%
[0.60, 0.65)	[4.20, 4.30)	6.1804e-01	1.3027e-01	21.08%
[0.60, 0.65)	[4.30, 4.40)	5.8084e-01	1.0966e-01	18.88%
[0.60, 0.65)	[4.40, 4.50)	6.1373e-01	9.9006e-02	16.13%
[0.60, 0.65)	[4.50, 4.60)	4.3586e-01	1.2932e-01	29.67%
[0.60, 0.65)	[4.60, 4.70)	5.2194e-01	2.1303e-02	4.08%
[0.60, 0.65)	[4.70, 4.80)	4.0951e-01	5.3210e-02	12.99%
[0.60, 0.65)	[4.80, 4.90)	2.6064e-01	7.1840e-02	27.56%

Double Differential Systematics (xF and Mass) - LD2 (23/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.60, 0.65)	[5.00, 5.10)	2.3192e-01	9.7261e-02	41.94%
[0.60, 0.65)	[5.10, 5.20)	1.6081e-01	4.6902e-02	29.17%
[0.60, 0.65)	[5.20, 5.30)	2.0732e-01	5.3089e-02	25.61%
[0.60, 0.65)	[5.30, 5.40)	1.9099e-01	4.9570e-02	25.95%
[0.60, 0.65)	[5.60, 5.70)	9.5902e-02	3.6133e-02	37.68%
[0.60, 0.65)	[5.80, 5.90)	1.0239e-01	3.8115e-02	37.23%
[0.60, 0.65)	[6.10, 6.20)	6.5617e-02	3.0792e-02	46.93%
[0.60, 0.65)	[6.20, 6.30)	8.8226e-02	3.7089e-02	42.04%
[0.60, 0.65)	[6.70, 6.80)	2.9597e-02	7.6108e-03	25.71%
[0.60, 0.65)	[6.90, 7.00)	2.1959e-02	7.3674e-03	33.55%
[0.60, 0.65)	[7.50, 7.60)	1.1676e-01	2.8804e-02	24.67%

Double Differential Systematics (xF and Mass) - LD2 (24/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.65, 0.70)	[4.30, 4.40)	3.7168e-01	4.4519e-02	11.98%
[0.65, 0.70)	[4.40, 4.50)	3.8042e-01	1.6568e-01	43.55%
[0.65, 0.70)	[4.50, 4.60)	2.8462e-01	3.9329e-02	13.82%
[0.65, 0.70)	[4.70, 4.80)	2.4993e-01	6.0941e-02	24.38%
[0.65, 0.70)	[4.90, 5.00)	2.1145e-01	3.1973e-02	15.12%
[0.65, 0.70)	[5.00, 5.10)	1.2456e-01	4.7650e-02	38.26%
[0.65, 0.70)	[5.10, 5.20)	1.3169e-01	5.1842e-02	39.37%
[0.65, 0.70)	[5.20, 5.30)	1.0242e-01	4.5334e-02	44.26%
[0.65, 0.70)	[5.40, 5.50)	8.6380e-02	2.9271e-02	33.89%
[0.65, 0.70)	[5.50, 5.60)	1.0694e-01	5.0404e-02	47.13%
[0.65, 0.70)	[5.70, 5.80)	8.6093e-02	4.2623e-02	49.51%

Double Differential Systematics (xF and Mass) - LD2 (25/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.65, 0.70)	[6.00, 6.10)	7.6218e-02	1.4480e-02	19.00%
[0.65, 0.70)	[6.30, 6.40)	2.6882e-02	1.0918e-02	40.62%
[0.65, 0.70)	[6.40, 6.50)	4.0263e-02	1.3297e-02	33.02%
[0.65, 0.70)	[6.50, 6.60)	6.2950e-02	1.0500e-02	16.68%
[0.65, 0.70)	[6.80, 6.90)	2.5282e-02	4.1089e-04	1.63%
[0.70, 0.75)	[4.20, 4.30)	2.9991e-01	2.2844e-02	7.62%
[0.70, 0.75)	[4.30, 4.40)	1.9978e-01	8.0263e-02	40.18%
[0.70, 0.75)	[4.40, 4.50)	2.4525e-01	6.1978e-02	25.27%
[0.70, 0.75)	[4.50, 4.60)	1.2651e-01	4.3198e-02	34.15%
[0.70, 0.75)	[4.60, 4.70)	1.3638e-01	6.3237e-02	46.37%
[0.70, 0.75)	[4.70, 4.80)	1.8269e-01	8.0007e-02	43.79%

Double Differential Systematics (xF and Mass) - LD2 (26/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.70, 0.75)	[5.00, 5.10)	1.1655e-01	3.2826e-02	28.17%
[0.70, 0.75)	[5.20, 5.30)	6.6634e-02	2.0137e-02	30.22%
[0.70, 0.75)	[5.40, 5.50)	6.8079e-02	2.3081e-02	33.90%
[0.70, 0.75)	[5.50, 5.60)	6.5151e-02	1.1797e-02	18.11%
[0.70, 0.75)	[5.70, 5.80)	2.7624e-02	1.3082e-02	47.36%
[0.70, 0.75)	[5.80, 5.90)	6.9067e-02	2.2731e-02	32.91%
[0.70, 0.75)	[5.90, 6.00)	5.3196e-02	1.3212e-02	24.84%
[0.70, 0.75)	[6.40, 6.50)	2.0408e-02	4.2582e-03	20.87%
[0.70, 0.75)	[6.50, 6.60)	1.8066e-02	3.2745e-04	1.81%
[0.75, 0.80)	[4.20, 4.30)	1.6560e-01	7.2220e-02	43.61%
[0.75, 0.80)	[4.40, 4.50)	8.2865e-02	2.7248e-02	32.88%

Double Differential Systematics (xF and Mass) - LD2 (27/27)

xF Range	Mass Range [GeV]	Mean σ_w	Std Dev $\sigma_{sys,w}$	Relative Error
[0.75, 0.80)	[4.80, 4.90)	7.5787e-02	1.9874e-02	26.22%
[0.75, 0.80)	[5.40, 5.50)	8.1148e-02	2.9476e-02	36.32%
[0.75, 0.80)	[5.80, 5.90)	3.9245e-02	5.9619e-05	0.15%
[0.75, 0.80)	[5.90, 6.00)	3.7811e-02	3.7342e-03	9.88%
[0.75, 0.80)	[6.10, 6.20)	2.3961e-02	1.0496e-02	43.80%

Why is the relative error still $\sim 13\text{--}18\%$?

- 1 **Statistical Noise Limitations:** While inverse-variance weighting heavily suppresses the noise of low-POT datasets (RS59, RS70), it cannot completely eliminate the pull from extreme outliers.
- 2 **Global Acceptance Mismatch:** A single `acceptance_mass_xF.root` file is currently used across all roadsets. Because different triggers have distinct kinematic acceptances, dividing by a global average creates artificial variance.

Path Forward: Implement roadset-specific Monte Carlo acceptance maps and evaluate the necessity of excluding extremely low-POT datasets from the overall variance calculation.